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From this assignment, I learned how to configure and verify Multi-Area OSPFv2 in a network. I understood that OSPF can divide a large network into different areas to improve routing efficiency and reduce the amount of routing information that routers need to process. Area 0 is the backbone area, and other areas such as Area 1 and Area 2 connect through it. I learned how to configure OSPF using a process ID, router ID, network statements, and wildcard masks. I also learned how Area Border Routers (ABRs) connect different OSPF areas. Using commands such as `show ip ospf neighbor`, `show ip route`, `show ip ospf database`, and `show ip ospf interface` helped me verify OSPF operation and troubleshoot connectivity. Finally, I learned how OSPF neighbors form a FULL adjacency and exchange routing information.

Reflection Questions

1. Which router(s) are internal routers?

R2 is an internal router because all of its interfaces belong to Area 0.

Note: In the full Packet Tracer topology, routers 4, 5, 6, 7, and 9 are also internal routers within their respective areas.

2. Which router(s) are backbone routers?

R2 is the backbone router because all of its OSPF interfaces are in Area 0.

3. Which router(s) are Area Border Routers (ABRs)?

R1 and R3 are ABRs.

- R1 connects Area 1 ↔ Area 0
- R3 connects Area 2 ↔ Area 0

4. Which router(s) are Autonomous System Boundary Routers (ASBRs)?

None in this topology. There is no router redistributing routes from another routing protocol or external network into OSPF.

5. Which routers are generating Type 1 LSAs?

All OSPF routers generate Type 1 (Router) LSAs for the areas in which they have OSPF interfaces.

6. Which routers are generating Type 2 LSAs?

Answer: The Designated Routers (DRs) on broadcast networks generate Type 2 (Network) LSAs. From your output, examples include routers 4, 5, 6, 7, and 9.

7. Which routers are generating Type 3 LSAs?

The ABRs R1 and R3 generate Type 3 (Summary) LSAs to advertise networks between OSPF areas.

8. Which routers are generating Type 4 and 5 LSAs?

No routers are generating Type 4 or Type 5 LSAs in this topology because there is no ASBR or external OSPF route.

9. How many inter-area routes does each router have?

From the routing tables you provided:

- R1: 0 inter-area routes shown in its routing table
- R2: 2 inter-area routes — 192.168.1.0/24 and 192.168.2.0/24
- R3: 1 inter-area route shown — 192.168.10.0/30

10. Why would there usually be an ASBR in this type of network?

An ASBR is normally used to connect OSPF to an external network or another routing protocol, such as EIGRP, RIP, BGP, or static routes. It redistributes those external routes into OSPF so OSPF routers can reach them.

The screenshot displays a Cisco Packet Tracer interface with a multi-area OSPFv2 network topology and a configuration window for router R1.

Topology:

- Area 0 (Yellow):** Contains routers R2 and R3. R2 has interfaces 10.2.1.0/24 and 192.168.10.0/30. R3 has interfaces 192.168.10.4/30 and 192.168.2.0/24.
- Area 1 (Pink):** Contains router R1. R1 has interfaces 10.1.2.0/24 and 10.1.1.0/24.
- Area 2 (Blue):** Contains router R3. R3 has interfaces 192.168.1.0/24 and 192.168.2.0/24.

Configuration Window (R1):

```
R1(config-router)# network 192.168.10.0 0.0.0.3 area 0
```

Step 3: Configure OSPFv2 on R2 and R3.

Repeat the steps above for R2 and R3 using a router ID of 2.2.2.2 and 3.3.3.3, respectively.

Part 2: Verify and Examine Multiarea OSPFv2

Step 1: Verify connectivity to each of the OSPFv2 areas.

From R1, ping each of the following remote devices in area 0 and area 2: 192.168.1.2, 192.168.2.2, and 10.2.1.2.

Step 2: Use show commands to examine the current OSPFv2 operations.

Use the following commands to gather information about your OSPFv2 multiarea implementation.

```
show ip protocols
show ip route
show ip ospf database
show ip ospf interface
show ip ospf neighbor
```

Reflection Questions

1. Which router(s) are internal routers?
2. Which router(s) are backbone routers?
3. Which router(s) are area border routers?
4. Which router(s) are autonomous system routers?
5. Which routers are generating Type 1 LSAs?
6. Which routers are generating Type 2 LSAs?
7. Which routers are generating Type 3 LSAs?
8. Which routers are generating Type 4 and 5 LSAs?
9. How many inter area routes does each router have?
10. Why would there usually be an ASBR in this type of network?

Suggested Scoring Rubric

Packet Tracer scores 80 points. Each of the Reflection Questions is worth 2 points.

Time Elapsed: 00:36:38 Completion: 80/80

Buttons: Top, Dock, Check Results, Back, 1/1, Next